

A project 2 Post 4 (Q/A)

① what are supervised learning algorithms

- Supervised learning algorithms are ML algorithms that learn from a labeled data, where results are already known and we choose, SLR, MLR, poly, and gradient function on them

② Diff. Between Classification & Regression

- Classification Model only predict the categories
For eg: (yes/no) for Bank loan can be given or not
while Regression work on continuous number
eg: price prediction

③ Explain SLR (Simple linear Regression)

- Simple linear regression is method used to model the relationship between one independent variable

④ List and explain the Assumptions of SLR

- (i) Linearity - independent & dependent variables should have a linear relationship
- (ii) Independence - observations should be independent for each other

(iii) Homoscedasticity - Error variance should remain constant

(iv) Normality of Residuals - Residuals follow a normal distribution

⑤ What is Bias-variance Trade-off?

→ The Bias-variance Trade-off describes the balance between underfitting and overfitting

⑥ Explain overfitting and underfitting

→ Underfitting occurs when model is too simple to capture.

Overfitting occurs when model learns noise and details from the training data